113A Introduction to Computer and Computer Science

Homework Assignment #6

Due: 11/4 12:00:00

In this homework assignment, you will practice how to apply the concept of object-oriented programming to solve a practical issue. As you have learned all the related knowledge during the lectures, you should have the ability to accomplish this.

Problem #1: Numerals

Numerals are one of the most important symbols in the history of mankind. They have been widely used in many parts of your life such as currency systems, mathematics, religions, …. About 2000 years ago, the Roman Empire had their numeral system to represent numbers. Unlike the current numeral system based on Arabic numerals, the system of Roman numerals uses special characters:

Symbol	Value
I	1
V	5
X	10
L	50
C	100
D	500
M	1000

In this homework, we are going to develop a program, which aims to transform an arbitrary Arabic numeral into Roman numeral. For simplicity, we only consider positive integers less than 4000 in this homework. In addition, you have to create some rules to tell Python how to deal with the operations like addition, subtraction and comparison of Roman numerals. You will need to use "class" to accomplish this task.

Please create a ***class object*** called ***numerals*** which can take **Arabic (int)** or **Roman numerals (str)** as input and meet the specification as shown below:

1. Show **Roman numerals** when you ***print***.
2. Follow the **Roman numerals** rules:
   (1) If one or more symbols are placed after another letter of greater value, add amount, vice versa.

(2) The value of the symbol is added to itself, as many times as it is repeated.
Ex. $\text{II} = 2,\ \text{III} = 3$.

(3) Symbols V, L, D will never repeat.
Ex. $\text{XV} = 15$, not VVV.

(4) The maximum repeated times of a symbol is three.
Ex. $\text{IV} = 4$, not IIII.

(5) Subtraction only uses symbols I, X and C.
Ex. $\text{XLV} = 45$, not VL.

(6) Subtraction only uses symbol nearby.
Ex. $\text{XCIX} = 99 \ldots (100 - 10 + 10 - 1)$, not $\text{IC} \ldots (100 - 1)$.

3. It can be compared $(>, <, \leq, \geq, =)$.
4. It can be operated $(+,\ -,\ \times)$.
5. Show the corresponding message if the number is less equal than 0 or greater equal than 4000.

Here is the sample code:

```python
class numerals(object):
    """ A class object for numerals """

    def __init__(self, value):
        """ Initialization """
        # 厂厂

        if isinstance(value, int) == True:
            # 厂厂

        elif isinstance(value, str) == True:
            # 厂厂

        else:
            # 厂厂
```

Some testing cases for you:

```
1.  print(numerals(345))
    CCCXLV
2.  print(numerals(987))
    CMLXXXVII
3.  print(numerals(345) > numerals(987))
    False
4.  print(numerals(345) + numerals(987))
    MCCCXXXII
5.  print(numerals(987) - numerals(345))
    DCXLII
6.  print(numerals("DDD"))
    ValueError: Wrong Roman numerals formation
7.  print(numerals(94) + numerals("LXXXVII"))
    CLXXXI
8.  print(numerals(9487))
    ValueError: Number out of range
```

Please accomplish this homework with an organized code (e.g., with main script and function script). Your main script file name must be 'main_hw6.py.' Here is a template for your code structure:

```
    113A_hw6_0123456789
          ├ ooo.py            # Module 1 for hw6
          ├ ⋮                 # Other modules if necessary
          └ main_hw6.py       # Main script of hw6
```

FRIENDLY REMINDER

You don't need to follow this structure exactly, just keep your main script clean.

Here is a template for your main script.

```python
if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="2024 NYCUDOPCS Homework #6")
    parser.add_argument("--n1", default=345, help="Testing number 1 (default: 345)")
    parser.add_argument("--n2", default=987, help="Testing number 2 (default: 987)")


    opt = parser.parse_args()
    n1 = numerals(opt.n1)
    n2 = numerals(opt.n2)


    # print the corresponding result
    print("Test cases:")
    print("="*60)
    print("    1. n1 =", n1)
    print("    2. n2 =", n2)
    print("    3. n1 > n2:", n1 > n2)
    print("    4. n1 + n2:", n1 + n2)
    print("    5. n2 - n1:", n2 - n1)
```

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
"""
Created on Wed Aug 7 01:23:45 2024

@author: Xi Winnie, student ID

@collaborators: William Lai, his student ID
"""
```

Please save your homework as a "**.zip**", "**.7z**", or "**.rar**" file, where the file name should follow this format:

113A_hw6_ID.zip

For example,

113A_hw6_0123456789.zip

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature in main script**. Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the "E3@NYCU" system. There will be a section for uploading your homework.